GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Plastics Engineering

Subject Code: DI05023031

Subject Name: Advanced Plastic Processing Techniques

w. e. f. Academic Year:	2026-27
Semester:	5th
Category of the Course:	PEC-III

Prerequisite:	Knowledge of different plastic materials and its conventional processing techniques
Rationale:	Advancements across various processing domains have significantly influenced the production capacity and efficiency of manufactured products. Similar progress has been observed in the field of plastic processing, where continuous innovations have enhanced both the quality and scalability of production. In particular, modern developments in key processing techniques such as injection moulding, extrusion moulding, and blow moulding have become essential areas of study. It is therefore imperative for final-year students to acquire updated knowledge of these advanced methodologies. Accordingly, this course is specifically designed to provide a comprehensive understanding of recent advancements in plastic processing techniques and their industrial applications.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Recall the principles, materials, and applications of advanced injection moulding, extrusion, and blow moulding processes.	R
02	Explain the working mechanisms and process variables of reaction injection moulding (RIM) and advanced extrusion techniques.	U
03	Classify different advanced plastic processing techniques based on their characteristics and applications	U
04	Apply appropriate processing parameters for injection moulding, extrusion, and blow moulding to produce given plastic products.	A
05	Select suitable plastic processing methods for manufacturing specific products based on material properties and design requirements.	A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
1.	**Advanced Injection Molding Processes** Material, process, advantages and disadvantages & applications of the following processes: 1.1 Gas-assisted injection molding 1.2 Sandwich injection molding 1.3 Structural foam injection molding 1.4 Flow molding 1.5 Metal filled injection molding 1.6 Multi color molding 1.7 Injection molding of reinforced thermoplastics	12	28% (20 Marks)
2.	**Reaction Injection Molding (RIM)** 2.1 Introduction to RIM process 2.2 Materials and additives 2.3 RIM process and variables 2.4 Flow diagram of RIM process 2.5 Machine & auxiliary equipment 2.6 Merits and demerits of RIM process 2.7 Applications of RIM	08	14% (10 Marks)
3.	**Advanced Extrusion Process & Machines** 3.1 Co- Extrusion 3.1.1 Co-extrusion structures 3.1.2 Barrier materials & adhesives 3.1.3 Comparison: Feed block die and multi manifold die 3.1.4 Advantages of co-extrusion products. 3.1.5 Applications of Co-Extrusion 3.2 Advanced Extruder Machine 3.2.1 Twin screw Extruder	08	20% (14 Marks)

	3.2.2 Intermeshing and no Intermeshing 3.2.3 Counter rotating and co-rotating 3.2.4 Comparison with single screw 3.3 Vented Screw Extruder 3.4 Internal Bubble cooling		
4.	**Specialized Extrusion Processes** 4.1 Nylon braided pipes 4.2 Hose pipe 4.3 Fishing net 4.4 Heat shrink film 4.5 Cling film 4.6 Corrugated sheets and pipes	07	15% (10 Marks)
5.	**Advanced Blow Molding Processes** 5.1 Stretch Blow Molding 5.1.1 Introduction 5.1.2 Single stage & two stage processes and its comparison 5.1.3 Orientation and stretch ratio 5.1.4 Pre-forming 5.1.5 Extrusion stretch blow molding 5.1.6 Injection orientation blow molding 5.2 Co-Extrusion Blow Molding 5.2.1 Co-extrusion equipment 5.2.2 Process 5.3 Miscellaneous Blow Molding Processes 5.3.1 Neck ring process 5.3.2 Drape process 5.3.3 Dip/displacement processes 5.3.4 Blow molding of irregular shaped parts	10	23% (16 Marks)
	Total	**45**	**100% (70 Marks)**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
19	**47**	**04**	**------**	**------**	**------**

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

Sr. No.	Title of Book	Author	Publication with place, year and ISBN
1.	Injection molding handbook	Dominick V. Rosato, Donald V. Rosato, Marlene G. Rosato	Publication: Springer Science & Business Media, LLC Publication Year:2000 ISBN: 978-1-4613-7077-2, 978-14615-4597-2
2.	Reaction injection moulding	Walter E. Becker	Publication: DA Information Services Publication Year: 1979 ISBN: 0442206313, 978-0442206314
3.	Injection Molding Alternatives: A Guide for Designers and Product Engineers	Jack Avery	Publication: Hanser Pub Inc Publication Year: 1998 ISBN: 1569902518, 978-1569902516
4.	Plastics Extrusion technology handbook	Sidney Levy & James F. Curley	Publication: Industrial Press Inc., Publication Year: 1989 ISBN: 0-8311-1185-2
5.	Extrusion of plastics	E G Fisher	Publication: Plastics and Rubber Institute Publication Year: 1976 ISBN: 0470150122, 9780470150122
6.	Plastic Extrusion Technology	Hensen Friedhelm	Publication: Hanser Gardener Publishers, NewYork Publication Year: 1989 ISBN: 9780195207606
7.	Blow Moulding of Plastics	E G Fisher	Publication: Butterworth-Heinemann Publication Year: 1971 ISBN: 9780592054384
8.	Plastic Blow Moulding Handbook	N.C. Lee	Publication: Springer Dordrecht Publication Year: 2012 ISBN: 9789401169905

(b) Open-source software and website:

1. https://www.bpf.co.uk/plastipedia/processes/Default.aspx#injectionmouldinggasassisted
2. https://www.bpf.co.uk/plastipedia/processes/Default.aspx#injectionstretchblowmoulding
3. https://www.hanserpublications.com/SampleChapters/9781569906033_9781569906033_LPR.pdf
4. https://www.scribd.com/presentation/481496088/Advanced-Injection-Moulding
5. https://www.youtube.com/watch?v=pcwyCvwDp1U
6. https://www.youtube.com/watch?v=so9OyGGlCv4
7. https://www.youtube.com/watch?v=AAvOYz7qOsw
8. https://www.youtube.com/watch?v=vGhGxuAGOIo
9. https://www.youtube.com/watch?v=qtweiXnnFMg

Suggested Course Practical List:

Sr. No.	Practical Outcomes (PrOs)	Unit No.	Approx. Hrs. required
1	Study the processing parameters & Operation of gas assisted injection moulding machine for given product.	I	04
2	Study the processing parameters, mould setting and Operation of sandwich injection moulding machine for given product.	I	04
3	Study the operation of multi-colour injection moulding machine for given product.	II	04
4	Study the process parameters and mould setting for given product of RIM process.	II	02
5	Study the operation and process parameters of co-extrusion multi-layer film plant.	II	04
6	Study the operation of twin-screw Extruder for PVC pipe plant.	III	02
7	Study manufacturing of Nylon braided pipes.	III	02
8	Study manufacturing of Corrugated pipes.	III	02
9	Study operation and processing parameters of injection stretch blow moulding machine.	IV	04
10	Study Neck ring process.	IV	02

List of Laboratory/Learning Resources Required:

Sr. No.	Equipment Name with Broad Specifications
1	Gas assisted injection moulding machine
2	Sandwich injection moulding machine
3	Multi colour injection moulding machine
4	Reaction Injection moulding machine
5	Multi-Layer Film Plant
6	Twin Screw Extruder
7	Injection stretch blow moulding machine
8	Extruder
9	Internal Bubble Cooling Ring mechanism

Suggested Project List:

a) Advanced Plastic Processes Reference Chart: Prepare a comprehensive technical chart comparing various advanced processes like Gas-assisted, Sandwich, and Structural foam injection molding.
b) Advanced Injection Molded Product Collection: Collect and categorize physical samples of products made using specialized injection molding techniques (e.g., multi-color or metal-filled) and analyze their specific design features.
c) Specialized Extrusion Product Portfolio: Create a display and technical report on products manufactured via specialized extrusion, such as nylon braided pipes, fishing nets, or corrugated sheets.
d) Advanced Blow Molding Sample Study: Collect and present various blow-molded articles, identifying the specific process used (e.g., stretch blow molding or co-extrusion blow molding) for each.
e) Sandwich Injection Mold Modeling: Design and create a physical or digital model of a sandwich injection mold to demonstrate the simultaneous injection of different materials.
f) Reaction Injection Molding (RIM) Process Model: Build a functional or illustrative model showing the RIM process, including the mixing head and mold components.
g) Process Variable Optimization Report: Conduct research and prepare a detailed report on how critical variables (like gas pressure in GAIM or stretch ratios in blow molding) affect final product quality.
h) Advanced Product Display Gallery: Organize a formal display of diverse plastic products created through advanced processing, labeling each with its material and manufacturing technique.
i) Environmental Impact & Recycling Study: Analyze the recyclability of multi-layer co-extruded films or sandwich-molded parts and suggest "green" manufacturing improvements.

j) Troubleshooting Guide for Advanced Molding: Create a manual identifying common defects in specialized processes like Structural Foam Molding or RIM, along with their root causes and solutions.

Suggested Activities for Students:

1. Collect various plastic articles from daily life and conduct a technical analysis to determine which advanced process was used for their manufacture.
2. Visit a nearby plastic processing industry specifically utilizing advanced techniques like RIM, co-extrusion, or specialized injection molding.3. Students will prepare identification chart with recycling codes.
3. Watch and analyze specialized video or animation films focusing on Advanced Injection Molding, RIM, and Advanced Extrusion.
4. Conduct a study on the environmental impact of advanced processes and propose methods to address sustainability and reduce waste.
5. Enroll in and complete modules from relevant Massive Open Online Courses related to specific sub-topics of advanced processing.
6. Select one of the "simpler or descriptive" topics from the syllabus (approx. 20% of the course) for self-study and present the findings to the class.
7. Prepare a report on the safety protocols practices required when operating high-pressure advanced molding machinery.

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